

Table 13.5 Trading Simulation Performance Measures

Performance Measure	Description
% Losing Trades	$LT = 100 \cdot \frac{\sum_{t=1}^N G_t}{NT}$ where $G_t = 1$ if Transaction Profit_t ≤ 0
Number of Up Periods	$N_{up} = \text{Number of } R_t > 0$ (13.46)
Number of Down Periods	$N_{down} = \text{Number of } R_t < 0$ (13.47)
Number of Transactions	$NT = \sum_{t=1}^N L_t$ where $L_t = 1$ if Trading Signal_t ≠ Trading Signal_{t-1}
Total Trading Days	$\sum (R_t)$ (13.49)
Avg. Gain in Up Periods	$AG = \left(\sum (R_t > 0) \right) / N_{up}$ (13.50)
Avg. Loss in Down Periods	$AL = \left(\sum (R_t < 0) \right) / N_{down}$ (13.51)
Avg. Gain/Loss Ratio	$GL = AG / AL$ (13.52)
Source: Dunis et al. (2004)	

Table 13.6 Trading Simulation Performance Measures

Performance Measure	Description
Probability of 10% Loss	$PoL = \left[\frac{(1-P)^{\left(\frac{MaxRisk}{\lambda}\right)}}{P} \right]$ where $P = 0.5 \cdot \left(1 + \left(\frac{((WT \cdot AG) + (LT \cdot AL))}{\sqrt{[(WT \cdot AG^2) + (LT \cdot AL^2)]}} \right) \right)$ and $\lambda = \sqrt{[(WT \cdot AG^2) + (LT \cdot AL^2)]}$ MaxRisk is the risk level defined by the user.
Profits T-statistics	$T\text{-statistics} = \sqrt{N} \cdot \frac{R^A}{\sigma^A}$
Source: Dunis et al. (2004)	